



What is trigonometry?

TRIGONOMETRY DEALS WITH THE RELATIONSHIPS BETWEEN THE SIZES OF ANGLES AND LENGTHS OF SIDES IN TRIANGLES.

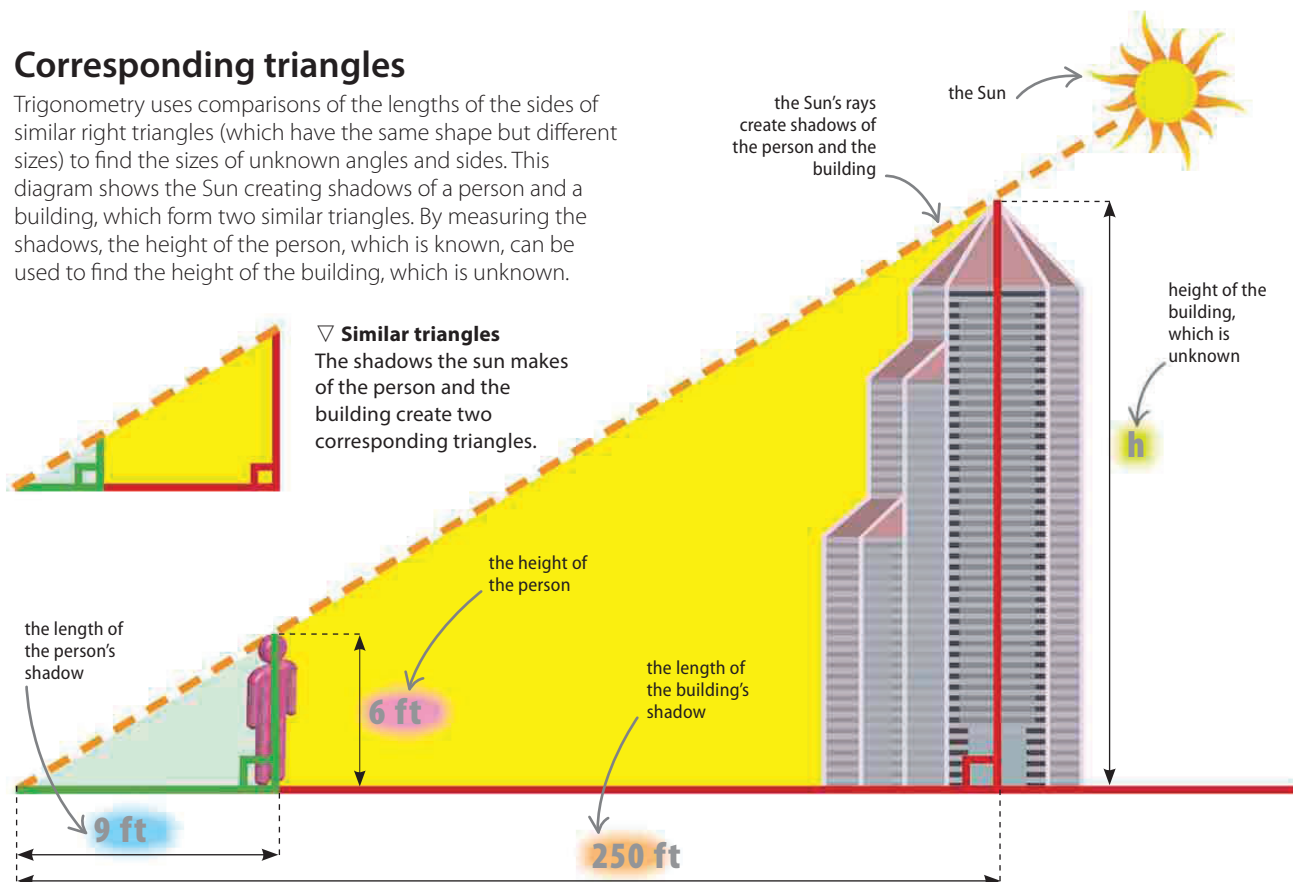
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◀ 56–59 Ratio and proportion

◀ 125–127 Similar triangles

Corresponding triangles

Trigonometry uses comparisons of the lengths of the sides of similar right triangles (which have the same shape but different sizes) to find the sizes of unknown angles and sides. This diagram shows the Sun creating shadows of a person and a building, which form two similar triangles. By measuring the shadows, the height of the person, which is known, can be used to find the height of the building, which is unknown.



▷ **The ratio between** corresponding sides of similar triangles is equal, so the building's height divided by the person's height equals the length of the building's shadow divided by the length of the person's shadow.

▷ **Substitute the values** from the diagram into this equation. This leaves only one unknown—the height of the building (h)—which is found by rearranging the equation.

▷ **Rearrange the equation** to leave h (the height of the building) on its own. This is done by multiplying both sides of the equation by 6, then canceling out the two 6s on the left side, leaving just h .

▷ **Work out the right side** of the equation to find the value of h , which is the height of the building.

$$\frac{\text{height of building}}{\text{height of person}} = \frac{\text{length of building's shadow}}{\text{length of person's shadow}}$$

$$\frac{h}{6} = \frac{250}{9}$$

the value of h is unknown

$$h = \frac{250}{9} \times 6$$

this side has been multiplied by 6 to cancel out the $\div 6$ and isolate h

whatever is done to one side of the equation must be done to the other, so this side must also be multiplied by 6

the answer is rounded to 2 decimal places

$$h = 166.67 \text{ ft}$$